Roll No. .....

# MAI–204

## M. B. A. (SECOND SEMESTER)

## END SEMESTER EXAMINATION, July/Aug., 2022

### STATISTICS FOR BUSINESS DECISIONS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) This question paper contains two Sections-Section A and B.

(ii) Both Sections are compulsory.

(iii) Answer any *two* questions among (a), (b) and (c) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 marks.

### Section—A

1. (a) Bring out the application of statistics in economics and business administration as a scientific tool. (CO3)

(b) 'Statistics is not a Science, it is a scientific method.' Critically explain and discuss the scope and limitations of statistics. (CO2)

*P. T. O.*

(c) On the basis of following data obtain regression of $y$ on $x$ and $x$ on $y$. (CO4)

| $x$ | $y$ |
|---|---|
| 15 | 12 |
| 27 | 15 |
| 27 | 15 |
| 30 | 18 |
| 38 | 22 |
| 46 | 26 |

2. (a) Explain clearly the meaning of Time Series Analysis. Indicate the importance of such analysis in business. (CO2)

(b) What is meant by measures of central tendency ? Discuss the merits and drawbacks of the various types of averages. (CO2)

(c) What is Regression ? Also define variability with the measures of variation. (CO2)

3. (a) Discuss different types of Distribution with special emphasis on Normal Distribution and its main characteristics. (CO2)

(b) Explain the following terms with illustrations : (CO2)

(i) One tail and two tail test

(ii) Type I and Type II error

(c) (i) Level of significance

(ii) Critical region. (CO2)

4. (a) What is hypothesis ? Explain errors in hypothesis testing. (CO2, CO1)

(b) Explain the relationship between ANOVA and t-test. (CO2, CO5)

(c) Write short notes on the following : (CO1)

(i) parametric statistics

(ii) Non-parametric statistics

(iii) F-test

## Section—B

5. **Case Study :** (CO6, CO4)

Formulate Hypothesis and critically examine and discuss below data. A sample of 200 persons with particular disease was selected. Out of these 100 were given drugs and others were not given any drug. The results are as follows :

| | Drug | No Drug | Total |
|---|---|---|---|
| Cured | 65 | 55 | 120 |
| Not Cured | 35 | 45 | 80 |
| Total | 100 | 100 | 200 |

Test whether drug is effective or note with Degree of Freedom of 1 and Chi-sq-5% 3.84.



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